Local ML Training Report (CIC-IDS-2017)

تقرير التدريب المحلي للموديل

Trained locally by: Mohamed Osama | Official ML owner: Habiba

Dataset: CIC-IDS-2017 MachineLearningCSV | Algorithm: Random Forest

Purpose: local playbook/attack-family alignment (team practice). Production ML API remains Habiba (H12).

Data summary

ملخص البيانات

Metric	Value
CSV files	8
Feature columns	74
Rows read	2,830,743
Kept after sampling	529,333
Train rows	372,295
Sampling caps	benign 50k/file, attack 30k/label
Hardware	Mac i7-9750H, 16GB RAM, n_jobs=4

Test metrics

مقاييس الاختبار

Metric	Value
Accuracy	0.9900
Weighted F1	0.9904
Macro F1	0.7391
Balanced accuracy	0.8479
Detection rate	0.9525
False-alarm rate	0.00306

Class notes

ملاحظات الفئات

- Strong: Benign, DDoS, PortScan, BruteForce, DoS
- Weaker (few samples): Bot ~F1 0.37, Infiltration ~F1 0.40

Artifacts

المخرجات

- models/ids_model.joblib (~1.9 MB)
- models/report.txt + models/data_audit.csv

Please sync feature column order and class labels with Habiba before detector integration.